

Twitch vs YouTube: Exploring how synchronicity is associated with social interaction and positive emotion among gaming video viewers

Seung Woo Chae^{a,*}, Mark Alberta^b, Sung Hyun Lee^c

^a Department of Journalism and Creative Media Industries, Texas Tech University, Lubbock, TX, USA

^b Department of Communication, University of the Pacific, Stockton, CA, USA

^c The Media School, Indiana University, Bloomington, IN, USA

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ABSTRACT

This study explores how synchronicity is associated with users' social interaction and positive emotion when watching gaming content on video-based social media platforms. YouTube and Twitch were selected as comparable venues wherein asynchronous and synchronous communication can be observed, respectively. To control the effect of video content, we found a gaming video on YouTube, the content of which had originally been streamed on Twitch. From the identical videos on the two platforms, the asynchronous comments on YouTube and the synchronous chat messages on Twitch were collected. We analyzed the two datasets using the text analysis program LIWC. The results showed that users are more likely to use social words in the asynchronous setting than in the synchronous setting. Meanwhile, positive emotion was more frequently observed in Twitch's synchronous texts compared to YouTube's asynchronous texts. These findings suggest the possibility that Twitch users are more like eSports spectators than chatters.

1. Introduction

Among the variety of social media platforms these days, video-based platforms such as YouTube, Twitch, and TikTok have attracted large numbers of users and distinguished themselves from other types of platforms, such as Twitter (primarily text-based) or Instagram (primarily image-based). A recent survey by the Pew Research Center revealed that the most widely used online platform among U.S. adults is YouTube, which had been used by 83 % of participants as of 2023 [1]. The survey also displayed the rising popularity of another video-based platform, TikTok; the percentage of participants who had ever used the platform increased from 21 % in 2021 to 33 % in 2023.

Current video-based platforms offer two main types of video services: live video streaming and video-on-demand (VoD). For the former, Twitch is one of the major platforms to watch live video-streaming broadcasts, and the service is accompanied by the functionality of internet relay chat (IRC), where users can communicate with the streamer or other viewers in real time. When it comes to VoD, YouTube is the largest UGC (user-generated content) VoD provider in the world. While the platform also supports its live streaming functionality, the fundamental service that has sustained the platform's popularity for about two decades is the untimed convenience and relative conciseness

of its VoD service [2]. Corresponding to the IRC of Twitch, YouTube allows users to post asynchronous comments under most videos on the platform so that users can give feedback to the uploader or express their opinion on the video at any time. Taken together, the major difference between the two types of services—live streaming and VoD—is whether the communication involves *synchronicity*, which refers to “a state in which actions move at the same rate and exactly together” [3, p. 581].

As we compared Twitch with YouTube, synchronicity can give different affordances to the platform users, which are not possible in an asynchronous setting, and the topic has come to the attention of many researchers who study computer-mediated communication (hereinafter CMC). As early as the 1980 s, Daft and Lengel [3] introduced media richness theory (MRT) and addressed how synchronicity affects the quality of interpersonal communication. Following that, Dennis and colleagues [4] indicated that MRT had some limitations in explaining communication in the CMC environment and suggested a new theory, media synchronicity theory (MST). Along the lines of those two theories, many studies have investigated how synchronicity affects the quality of communication in different environments [5,6]. To the best of our knowledge, there is not a single study that addresses how synchronicity affects communication in the context of watching gaming content or other entertaining content. It may be mainly because both MRT and MST

* Corresponding author at: 3003 15th St, Room 511, Lubbock, TX 79409, USA.

E-mail address: seuchae@ttu.edu (S.W. Chae).

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1875-9521/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

were initially designed to address communication in business settings and measure how synchronicity is associated with work performance [4,3]. Accordingly, many of the following studies based on those theories have applied the concept of synchronous communication in a business setting [5] or in other non-entertaining settings such as distance learning [6].

The current study fills this gap in the literature by exploring synchronous communication on video-based platforms when the video content is purely entertaining. Of particular interest to this study is how synchronicity affects users' social interaction and positive emotion when watching gaming content. Social interaction has been suggested by multiple studies as one of the major motivations why people visit those video-based social media platforms including Twitch [7] and YouTube [8]. While the findings showed that the users want to communicate with others there, it is still an understudied area how the social interaction varies by (a)synchronicity. Meanwhile, there are only a limited number of previous studies regarding how synchronicity is associated with users' positive emotion when watching gaming content. Given that the major reason that most people watch entertaining content including gaming content may be to experience positive emotion, we believe that positive emotion is a topic that deserves more attention from researchers. Taken together, it is our belief that investigating the associations between synchronicity and those two variables can contribute to a better understanding of how synchronicity is related to social media users' desires for social interaction and enjoyment.

In order to test the associations, the present study collects text data from synchronous chat messages on Twitch and asynchronous comments on YouTube, respectively. While the ideal research design could be collecting both synchronous and asynchronous data from a single platform to control the platform effect, in practice, many streamers prefer to stream on Twitch and upload a part of the video on the other platform YouTube as an edited clip. We recognize that this phenomenon itself is one that we should observe as it is, particularly since the present study is not an experiment focusing on causality by controlling confounding variables but rather an exploratory content analysis observing what practically happens in the real world.

2. Literature review

Despite the appearance of other video-based platforms, YouTube, which is one of the earliest video-based platforms launched in 2005, has retained its leading position and further kept expanding its user base since its beginning [9]. One of the challengers to YouTube, Twitch was introduced in 2011 and has distinguished its services by focusing on its live streaming service. The strategy has seemed effective so far; the platform has achieved, on average, more than 100,000 concurrent streamers and 2.4 million concurrent viewers (as of January 2025) through the last 14 years [10]. It is worth mentioning that YouTube began its first live streaming in 2008 [11], which is earlier than the launch of Twitch, but the company did not move its focus in that direction and kept the search-oriented VoD as its major service.

In what follows, the relevant literature exploring communication on each platform is introduced focusing on the commenting functionality, followed by the discussion of prior research on how particularly social interaction and positive emotion are associated with the (a)synchronous communication on the platforms.

2.1. Communication on YouTube

In accordance with its enormous popularity, numerous researchers have examined diverse communication phenomena around YouTube [12,13,14]. When YouTube was first introduced in 2005, the platform was considered unique because before the emergence of YouTube, text (or sometimes image) was employed as the major modality of most early social media platforms such as *SixDegrees.com* or *Friendster* [15]. Along with the trend, many of the early CMC researchers recognized text as a

key media form with which we can expand our understanding of social media and general CMC [16]. That being said, in 2005 when YouTube was first introduced, the platform pioneered a new type of *video-based* social media platform. The impetus that made the platform distinguishable from previous ones was not its advanced information processing technology per se but the then-fresh affordance that people can communicate with each other using videos, and the video-oriented service has still attracted a myriad of users worldwide [9]. It is worth highlighting that not only consuming but also producing videos became a new trend led by YouTube [2]. The platform has allowed ordinary people to upload their own videos—so-called UGC (user-generated content)—without hassle, and the videos contain “real” messages that people want to share, ranging in topic from funny jokes to social movements [17].

However, over time, researchers have realized that video is not the only modality that we need to pay attention to on YouTube. Another important communication dimension can be found in the comment thread area under the videos. Not long after the launch of the platform in 2005, Lange [18] notes the power of YouTube comments on social interaction saying, “intelligent commentary on a video could stimulate closer social interactions” (p. 376). Following the study, numerous studies have confirmed that comments are fundamental to making YouTube a venue for social interaction where people can express their thoughts on the videos, also see other users' thoughts, and finally communicate with each other [19,20,21]. Their discussions ranged over a variety of topics from social issues including sexism or racism [20] to international issues such as refugee issues [19] or anti-government protests [21].

Meanwhile, there are researchers who found that the comment functionality on YouTube has a certain limitation as a communication tool. Dynel [12] claims that communication on YouTube is much more complicated than communication on TV because it is composed of three different levels: 1) the speaker and hearers in video interaction, 2) the sender and the recipient of a YouTube video, and 3) speakers and hearers on comment threads who post and read comments. Building on Dynel's third level (i.e., communication in comment threads), Herring and Chae [22] define YouTube as a prompt-rich platform and examine if the addressees and topics in comment threads could be confused due to the complexity of the communication framework on YouTube. The results confirmed that comments on YouTube sometimes cause confusion about the topics and addressees in the conversation, and it could be a hurdle to coherent social interaction. Although there are multiple studies that say the complexity itself is one of the enjoyable points in the CMC environment [23], it must be also true that they sometimes work as a barrier against effective communication and social interaction [24].

It is worth noting that, while there has been a body of research exploring YouTube's commenting functionality, only a limited number of studies have focused on YouTube comments under gaming videos, which the current study tries to examine. This gap in the literature may be partially because those comments are recognized as socially less important than the comments on other videos about social or political issues. However, Google's report about YouTube trends in 2021 highlights the increased importance of the gaming content by saying, “[g]aming” used to be a subculture, but with 2.9 billion people expected to play video games by the end of this year [2021], today it's clearly pop culture” [25]. In line with the notion, we expect that the current study can extend the existing literature on YouTube by observing users' social interaction and emotion when watching gaming content.

2.2. Communication on Twitch

Communication on Twitch can be somewhat more complicated than that on YouTube since it is not only another prompt-rich platform but also the communication there is mostly synchronous. The synchronicity allows dynamic hybrid communication that involves multiple types of sub-communication, each of which is conducted via a different media

form. One of the early papers on Twitch by Hamilton and colleagues [26] separates the media types by referencing McLuhan's concepts of hot media and cool media [27]. The original author introduces the concept of *definition*, which refers to the extent of being filled with data, as a principle that distinguishes hot media and cool media. While hot media entail high definition eliciting the audience's low participation such as radio, cool media entail low definition eliciting the audience's high participation. Employing the notion, Hamilton et al. [26] describe Twitch as a hybrid media mixed with both hot and cool media, composed of game graphics (high fidelity), live webcam video (medium fidelity), and chat (low fidelity).¹ Along the lines of Hamilton et al.'s study [26], Sjöblom and Hamari [28] argue that "the mix of media allows for a highly interactive experience" (p. 986) since Twitch's IRCs can function as a venue for social interaction, and video content can serve as a facilitator for the interaction. Following those studies, several studies have explored Twitch users' social interaction in their chat messages until recently [29,30].

One distinct point in the literature on Twitch, compared with that on YouTube, is the fact that most of the Twitch studies have focused on gaming content [26,29]. Since Twitch originally came from a section of *Justin.tv* that particularly covered gaming content, the platform has been a playground of gamers from its early stage [26,31]. Over time, Twitch game streamers have earned rigid fandoms wherein their subscribers regularly visit their channels to watch the streamers' gaming content [32]. This concentration is substantially different from YouTube, which covers diverse topics such as education, politics, entertainment, etc. [2,17].

2.3. Social interaction and synchronicity

As stated, previous studies about both YouTube and Twitch indicate that social interaction through commenting/chatting is one of the important communication dimensions on those platforms [33]. Furthermore, the studies suggest an idea in common that a high proportion of their users visit the platforms again to engage in social interaction [7,8]. In other words, social interaction is not just an available activity that users who visit the platforms can enjoy but a critical motive for which the users come there.

An early study regarding communication on YouTube by Haridakis and Hanson [13] found that YouTube users have a distinct social motivation compared to TV viewers and argued that it reflects the social networking characteristics of YouTube. Following the study, multiple researchers have also demonstrated that social interaction is a salient motivation to visit YouTube, predominantly employing uses and gratification theory as their theoretical framework [8,34]. Khan [8] conducted a survey about YouTube users' motivation, and the results show that users' commenting behavior is particularly associated with their social motivation. The study reveals that, while there are other motivations stronger than social motivation regarding the general purpose of visiting YouTube, commenting behavior is most associated with social motivation over others. Likewise, there have been a number of studies arguing that commenting behavior is connected to the formation of online communities based on YouTube channels [35].

When it comes to Twitch, there are also numerous studies suggesting that the popularity of the platform is supported by social motivation [36]. Some of them delineate Twitch as a platform fostering participatory communities with the IRC feature, and a substantial proportion of which ultimately constitute fandoms of the game streamers [26,37]. Having said that, while the findings from previous studies about social interaction on YouTube have been quite consistent, some of the studies about social interaction on Twitch have shown inconsistent findings. In contrast to the strand of research that found the importance of social

interaction among Twitch users [36,37], some other researchers pointed out that massive chat messages in synchronous Twitch chats do not allow coherent interaction between users or between users and the streamer, which can lead to only a limited level of social interaction. Ford and colleagues [38] examine communication practices in a popular channel's IRC while spectating a game, *Hearthstone*. They concluded that normal and coherent communication cannot be achieved in the condition of Twitch chats, which is usually filled with a large scale and fast pace of chat messages. The authors described the communication there as a "crowdspeak" and argued that it is impossible to apply traditional CMC approaches to the synchronous communication on Twitch. Put differently, the authors do not recognize IRCs on popular Twitch streamers' channels as an ideal venue for social interaction.

We recognize that Ford et al.'s [38] finding deserves to be explored further by comparing Twitch's synchronous communication with YouTube's asynchronous communication. Thus, the current study measures the extent of social interaction in the chat messages from Twitch and that in the comments from YouTube and tests if there is a significant difference between them. Taken together, we propose the following hypothesis based on prior literature:

H1: Twitch's synchronous chat messages will show less social interaction than YouTube's asynchronous comments.

2.4. Positive emotion and synchronicity

Positive emotion is another essential variable when it comes to entertaining content on video-based platforms including gaming content. Given that many people who watch entertaining videos may ultimately want to earn positive emotion through their experiences, it deserves to be researched how synchronicity is associated with users' positive emotion. In fact, in the literature on communication on YouTube, there are a group of studies analyzing sentiment including positive emotion on YouTube comments, most of which used natural language processing (NLP) methods [39,40]. Nonetheless, since the previous studies on YouTube have rarely researched gaming videos, it is still an unexplored area how users' emotion is associated with gaming videos on YouTube. Meanwhile, in the literature on communication on Twitch, some studies have explored positive emotion in Twitch chat texts [29]. To our knowledge, none of them researched how synchronicity is associated with sentiment.

Although it is not related to either YouTube or Twitch, Park and Sundar's [41] study gave us an important clue regarding the association between synchronicity and positive emotion. In their study, the authors investigated how synchronicity can affect users' degree of positive emotion in a setting of communication with a live agent. The results displayed that the participants in the high synchronicity condition, which brought no delay in response time, showed higher positivity than other participants in the medium and low synchronicity conditions. Although the experiment was conducted to address consumers' satisfaction, which is a considerably different topic from the current study, there is a possibility that positivity is an essential effect that synchronicity can arouse regardless of the specific communication setting. Therefore, based on Park and Sundar's [41] findings, our second hypothesis is suggested as follows:

H2: Twitch's synchronous chat messages will show higher positive emotion than YouTube's asynchronous comments.

3. Methods

3.1. Data collection

To test how synchronicity is associated with users' social interaction and positive emotion, we first decided to control one crucial factor that can significantly affect both of the variables: the content of the video. It

¹ Hamilton et al. [26] employed the concept "fidelity" instead of McLuhan's [27] concept "definition."

is rather commonsensical that viewers' comments or chat messages vary by the game, the streamers' tone, or some other factors in the video content. Accordingly, cases that have the same video content on both Twitch and YouTube were searched for control. While the initial plan was to find a large number of cases to conduct random sampling, the plan was frustrated by a couple of practical hurdles. First, although we found multiple YouTubers/Twitch streamers who uploaded their streamed Twitch videos on their YouTube channels, we also found that many of the originally streamed videos on Twitch had been already removed. It was because of the archiving policy of Twitch: the platform automatically removes streamed videos broadcasted longer than 2 months ago whereas most YouTube videos are permanently preserved. Another more critical hurdle was that it was hard to find a YouTube video that was not edited from the originally streamed one. It turned out that many Twitch streamers upload their videos on their YouTube channels after cutting out some loose parts and editing them to fit into YouTube.

Adjusting to the situation, we instead decided to find a single best case based on our criteria: first, the videos on YouTube and Twitch are perfectly identical; second, the videos have enough numbers of YouTube comments and Twitch chat messages, respectively, so that they properly represent each of the two audience groups; third, the game in the videos is appropriate for users' social interaction. To be specific about the last one, it was considered that visually and audibly compelling games such as first-person shooter games are not appropriate for this study since the users can be overwhelmed by the features of the game without having chances to talk to each other.

Based on these criteria, videos of a Twitch streamer who also operates a YouTube channel under the same name were decided as data sources. According to Wikitubia, a website that provides information about YouTubers, the streamer is an American male and started his YouTube channel in 2015. On Twitch, he livestreams gaming content on most days as a professional streamer. Clips of his Twitch streams are frequently uploaded to his YouTube channel. He has about 933,000 followers on Twitch and 659,000 subscribers on YouTube as of October 2024. Of a variety of gaming videos on his channels, a video in which the streamer played the social game *Among Us* with nine other players in December 2020 was selected. *Among Us*, released in 2018, was one of the most popular games on Twitch in 2020 [42]. As the mission of the game can be achieved by social deduction, it was assumed that the game is favorable to social interaction between the viewers.

Comments under the YouTube video, which is 16 min and 33 s long, and the chat messages during the corresponding part of the Twitch video were collected separately. The YouTube comments were collected via the platform's API (Application Programming Interface) using the programming language Python. The Twitch chat messages were collected via Twitch Downloader, a software tool that allows the downloading of recent Twitch chat logs. All YouTube comments and Twitch chat messages were included in the final sample without any exclusions. The unit of analysis was each comment or chat message. The sample size was 2660 messages, composed of 1489 Twitch chat messages and 1171 YouTube comments.

3.2. Analysis methods

A number of previous studies analyzing texts on YouTube and Twitch have used Linguistic Inquiry and Word Count (commonly referred to as LIWC) as a computational text analysis program and confirmed its validity as a research tool to examine texts on those platforms [31,43]. Following the studies, the present study also employed the 2015 version of LIWC (hereinafter LIWC) as a text analysis tool. It is developed based on psychology and linguistics, and its basic mechanism operates by word count through its pre-existing dictionaries [44]. While LIWC has 93 dictionary categories, the current study uses the two relevant categories, social words and positive emotion [45]. The dictionary of social words is composed of 756 words including words regarding other people (e.g.,

friends, they) or other words pertaining to social interaction (e.g., Instagram, talk). LIWC reports that the internal reliability score of the category was 0.86 (corrected alpha [44]). Meanwhile, the dictionary of positive emotion has 620 words, and the internal reliability score was 0.64 (corrected alpha; Pennebaker et al., 2015). A score for each category shows an average number of words in 100 words that belong to the dictionary; in other words, the scale is equivalent to a percent probability [45].

Since the unit of analysis is an individual comment/chat message, the score of each variable for each platform is the average score of all the text data in each platform group. Results are presented by descriptive statistics, and *t*-tests are conducted to see the statistical significance of the results. In addition, the scores are compared to the reference data provided by LIWC to understand how the score of each group is positioned relative to other general types of texts (or speech [44]). In order to provide a sense of the degree to which language varies across communication types, LIWC collected reference texts from six different genres, which are expressive writing, natural speech, blogs, novels, *The New York Times*, and Twitter. Besides, there is a category referring to the shared means of the six genres called "grand means." For the references, LIWC collected a total of more than 231 million words across the genres [44].

To avoid any possible confusion, it is necessary to clarify that, while comparing values within the same dictionary category (i.e., social words or positive emotion) can be valid, comparing values across different LIWC categories is pointless because the number of words in each dictionary differs by category.

4. Results

4.1. Social words

For the total of 2660 texts including both Twitch chat messages and YouTube comments, the average score of social words was 7.92 ($SD = 17.10$, see Table 1 for the overall comparison). When it comes to each group, the YouTube group showed an average score of 10.47 ($SD = 15.05$), and it was about 77 % higher than that of the Twitch group, which scored an average score of 5.91 ($SD = 18.31$). A *t*-test confirmed that the difference between the two groups was statistically significant, $t(2652.9) = -7.05$, $p < 0.001$. Thus, the first hypothesis expecting that Twitch's synchronous texts show less social interaction than the other was supported.

Comparing the results to the references of LIWC (see Fig. 1), the social words score of the YouTube group (10.47) was higher than any of the references (Twitter: 10.47, Blogs: 8.95, Natural speech: 10.42, Expressive writing: 8.69, NY Times: 7.62, Grand mean: 9.74) except for Novels (12.26). It is worth mentioning that the score of the YouTube group was almost identical to that of another social media platform, Twitter; both of them showed 10.47 after rounding off. That being said, the score of the Twitch group (5.91) was higher than none of the references. It was even 22 % lower than the average score of *The New York Times* ($M = 7.62$), which was the lowest one in the reference group.

4.2. Positive emotion

The average score of positive emotion across all of the data was 12.36 ($SD = 26.38$, again see Table 1). The Twitch group showed an average

Table 1
Overall comparison between the groups.

	Social words***	Positive emotion***
Twitch	5.91	14.36
YouTube	10.47	9.83
Total	7.92	12.36

*** The *p*-value is less than 0.001.

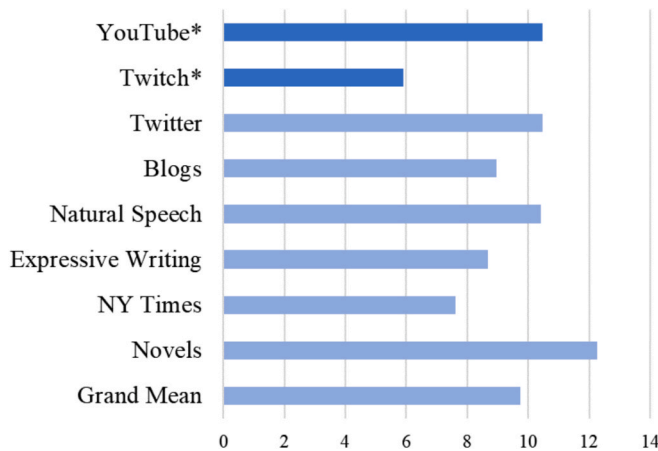


Fig. 1. Social words score by text category. * YouTube and Twitch represent the data of the current study, while the others are from the LIWC development manual [44].

score of 14.36 ($SD = 31.46$), and it was 46 % higher than that of the YouTube group, which recorded an average score of 9.83 ($SD = 17.67$). A t -test confirmed that the difference between the two groups was statistically significant, $t(2425.1) = 4.69, p < 0.001$. Therefore, the second hypothesis expecting that Twitch's synchronous texts show higher positive emotion than the other was supported.

When it comes to a comparison to the LIWC references (see Fig. 2), both of them are higher than any of the references (Twitter: 5.48, Blogs: 3.66, Natural speech: 5.31, Expressive writing: 2.57, NY Times: 2.32, Novels: 2.67, Grand mean: 3.67). The scores of Twitch and YouTube were 162 % and 79 % higher than the score of Twitter, respectively, which was the highest among the references.

5. Discussion

5.1. Social interaction

Our first hypothesis predicted that Twitch's synchronous chat messages show less social interaction than YouTube's asynchronous comments. Corresponding to that, the results displayed that the asynchronous comments on YouTube displayed nearly 1.8 times more social words than the other, and the difference was statistically significant. In fact, the social words score of the Twitch group was lower not only than the YouTube group but also than any of the references of LIWC

[44]. Given that the references include the category of natural speech, which represents our everyday language, it can be interpreted that the condition of the Twitch IRC was less favorable to social interaction than general conversations in our daily lives.

Although our finding about social interaction is consistent with Ford et al.'s study [38] specifically about social interaction on Twitch, it is worth reiterating that previous studies in general CMC research have dominantly suggested that social interaction is more active with synchronicity than without it [3,46]. It is mainly because the theories about synchronicity such as MST propose that interaction can be more effective as communication is more like *face-to-face* communication [4]. Likewise, in the studies drawing on those theories, face-to-face communication is often considered a somewhat ideal form of communication that CMC should try to resemble. Along with the notion, synchronicity—which enables instant responses like face-to-face communication—is generally assumed to be a favorable factor for a high quality of communication in the CMC environment.

However, the finding of the present study challenges this belief. Our results did not tell us that synchronicity is helpful for social interaction but rather showed a case that synchronicity can provide an unfavorable condition to social interaction compared to asynchronous communication. Although we cannot infer the precise reason behind the finding within the research design of this study, we can consider some possible rationales based on prior literature. The most possible reason is the overwhelming quantity and speed of the IRC messages. As aforementioned, Ford et al. [38] indicate this problem and argue that popular IRCs on Twitch are inappropriate for coherent conversation. This perspective is in accord with an earlier study on IRC [47] which suggests that 300 concurrent users in one chat channel and 30 chat messages per minute are cognitive limits for coherent communication. In practice, it is easy to find a channel on Twitch that has more than the suggested threshold for either number of concurrent users or number of messages per minute, and our Twitch data also had, on average, 89.97 messages per minute.

In fact, the finding that Twitch's synchronous chats are not favorable to social interaction seems somewhat commonsensical as well. It is true that there are a myriad of IRCs on Twitch wherein each user writes messages mostly about their feelings about the stream regardless of other users' thoughts, which sometimes end up being filled with emojis. Taken together, even if Twitch users have a desire for social interaction [36], popular streamers' synchronous IRCs may frustrate that desire with their "waterfall of text" [26].

5.2. Positive emotion

Our second hypothesis expected that Twitch's synchronous chat messages show higher positive emotion than YouTube's asynchronous comments. The results coincided with the hypothesis; the synchronous chat messages from Twitch displayed about 1.5 times higher positive emotion than the other, and the difference was statistically significant. Our results about positive emotion are consistent with the finding of Park and Sundar [41], which found that synchronicity can increase customers' positive emotion in the setting of communication with a live agent. Thus, there is a high possibility that the positive association between synchronicity and positive emotion is essential regardless of the type of communication.

It is worth noting additionally, while their scores were significantly different, both of the groups manifested higher positive emotion than any other reference categories of LIWC. In other words, both YouTube and Twitch effectively work as entertaining platforms wherein people can enjoy their services and share positive emotion.

5.3. Twitch IRC participation

The interpretation of the findings of this study can become rather more complicated when we put the finding about positive emotion

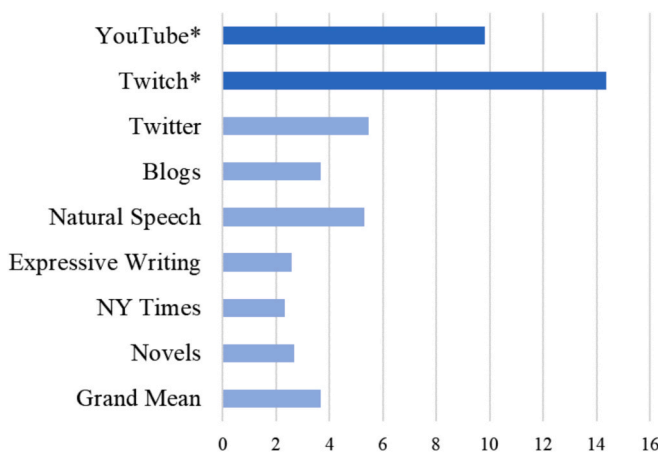


Fig. 2. Positive emotion score by text category. * YouTube and Twitch represent the data of the current study, while the others are from the LIWC development manual [44].

together with the finding about social interaction; synchronicity does not help users' social interaction, yet the users are more likely to feel positive emotion with synchronicity than without it. This can further lead us to a more fundamental question about Twitch users' IRC participation: What is the enjoyment of people participating in Twitch IRCs?

One possible answer to that inquiry comes from the notion that Twitch users may watch gaming content as eSports, which refers to "alternative sport realities, that is, to electronically extended athletes in digitally represented sporting worlds" [48, p. 199]. Many traditional sports fans enjoy expressing their comments and judgments on the game regardless of whether it can make social interaction or not; to illustrate, sports fans in the stands do not shout for interaction, but they enjoy their shouting itself. Likewise, it is possible that eSports viewers also feel positive emotion while expressing their ideas and feelings about the game play irrespective of reaction to the expression. Taken together, Twitch IRCs may be a venue where users can satisfy their desire to express their feelings about the video content rather than their desire for social interaction. In that sense, Twitch IRCs may be distinguishable from other ordinary IRCs since the users in Twitch IRCs are more like *spectators* than *chatters*.

Accordingly, the social motivation to visit Twitch needs to be revisited by future research. One possible research question is whether the social interaction that users want is interaction between users or between a user and the streamer. Particularly given that a number of recent studies on Twitch found parasocial relationships—the illusion of a "face-to-face relationship" between media personalities and their audience—between Twitch users and their streamers [31,43,49], it is plausible that the social interaction that the users want is that with their streamers. If so, the structure of the communities in Twitch may be similar to the structure of fandoms, which has nodes connecting each user and the streamer without necessary nodes between the users.

5.4. Limitations and future research

The current study has several limitations that should be acknowledged. First, because of its small sample size, the main thesis of this study is limited in scope. Again, we decided to select one particularly suitable case that would allow us to compare synchronous communication on Twitch and asynchronous communication on YouTube, due to Twitch's archiving policy. That said, although the generalizability of its findings is limited, the current study reveals a critical possibility that the general findings of the previous studies in CMC about synchronicity cannot be applied to certain CMC contexts in practice. We hope that future studies will be able to verify our findings using a larger sample size.

With the expanded sample, there is another important direction that future research can explore. As mentioned, the case that this study examined is drawn from a popular Twitch streamer's channel. However, there are a myriad of Twitch channels that have a small number of followers, which might fall below Jones et al.'s [47] threshold for either concurrent users (300 users) or chat messages per minute (30 messages). Looking into whether the texts from the small IRCs have more social words than those from large IRCs could be a meaningful topic for future research, as Hilvert-Bruce et al. [7] found that small channel viewers on Twitch have higher social motivation than large channel viewers. Furthermore, how the extents of social interaction and positive emotion vary by content type or the streamer's characteristic can be another topic that can be examined with a large sample.

Lastly, another noteworthy limitation originates from the intrinsic limitation of computational text analysis. Although the text analysis program LIWC has relatively reliable capabilities for analyzing texts. It is true that the mechanism still has a certain degree of limitations to interpret texts. In fact, Twitch is a place where users often use jargon or slang, which can sometimes only be interpreted within the specific community. Given that the dictionaries of LIWC were pre-tested by more

general texts than those on Twitch, a certain group of words that are only used in the Twitch community may have been interpreted incorrectly. We hope that future studies obtain more accurate results by employing their own dictionaries, created for specific types of channels or games.

6. Conclusion

In spite of these limitations, the current study provides new insights regarding how synchronicity is associated with users' social interaction and positive emotion on video-based social media platforms. Our results showed that YouTube users in the asynchronous setting were about 1.8 times more likely to employ social words than Twitch users in the synchronous setting. This corresponds to Ford et al.'s [38] findings that popular Twitch IRCs are not ideal environments for social interaction because of the overwhelming volume and speed of chat messages. Meanwhile, positive emotion was about 1.5 times more frequently found in Twitch's synchronous chat messages than YouTube's asynchronous comments. These results align with Park and Sundar's [41] finding that synchronicity can increase customers' positive emotion in the setting of communication with a live agent. Taken together, our findings suggest the possibility that Twitch users watch gaming content as eSports and are more like shouting spectators in the stands than casual chatters, allowing them to experience positive emotion without engaging in social interaction. We hope future research can further examine this possibility and expand our knowledge about synchronicity in CMC.

CRedit authorship contribution statement

Seung Woo Chae: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Mark Alberta:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Sung Hyun Lee:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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